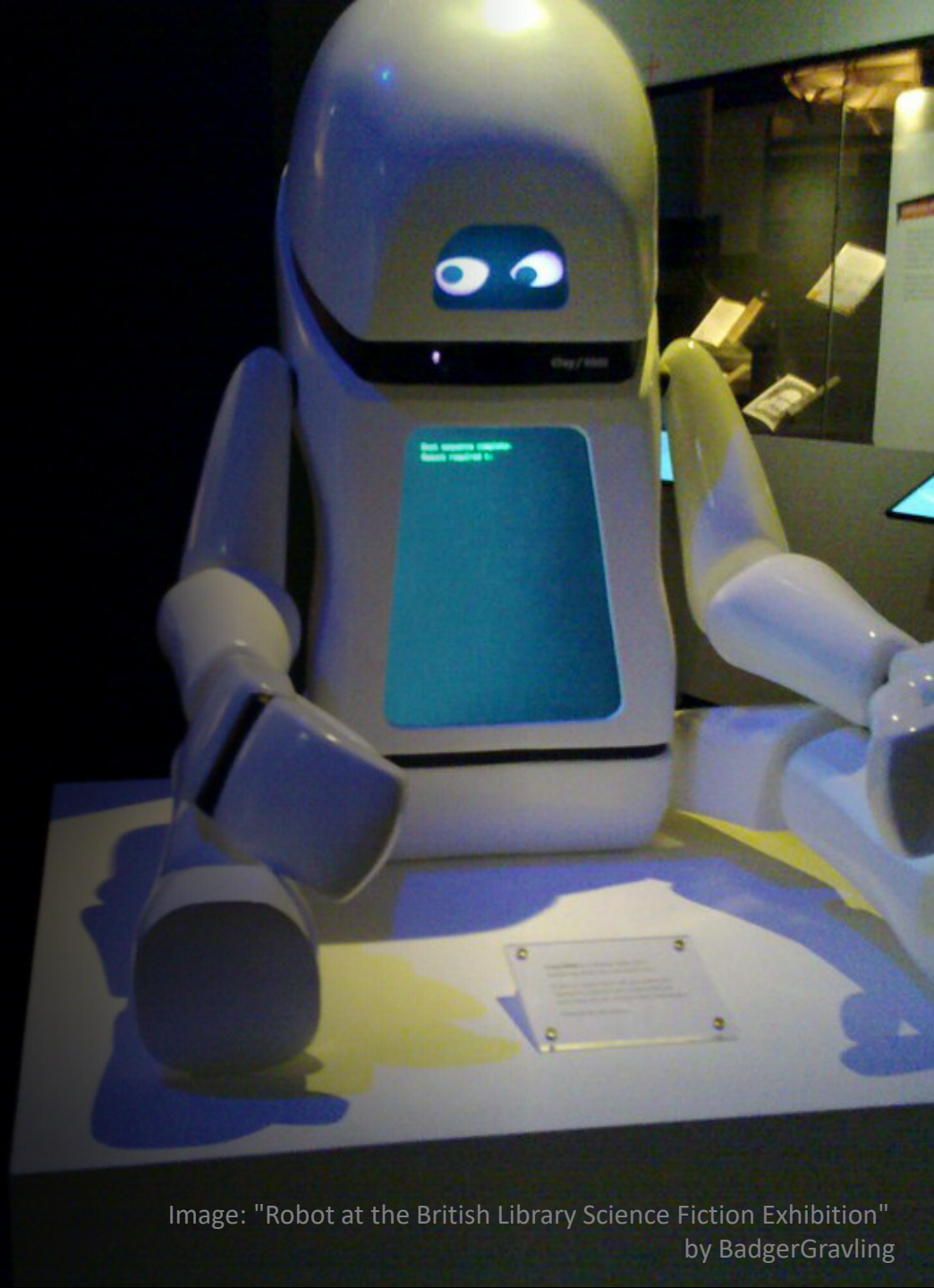


Discussion

CS 5/7320
Artificial
Intelligence

Intelligent Agents
AIMA Chapter 2

Slides by Michael Hahsler
with figures from the AIMA textbook.



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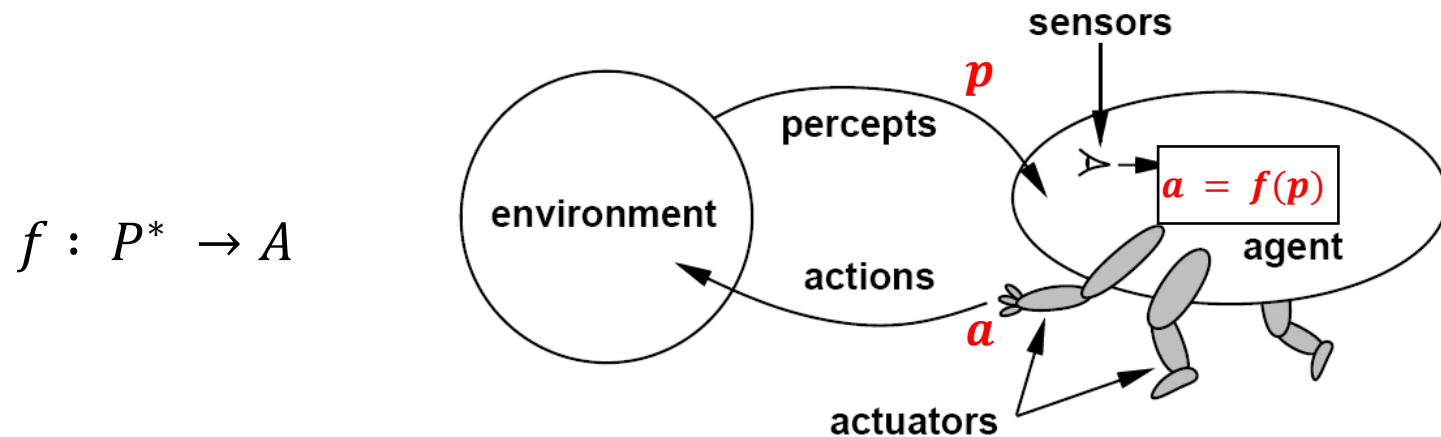
Image: "Robot at the British Library Science Fiction Exhibition"
by BadgerGravling

Module Review 1



Agent Function and Agent Program

The **agent function** maps from the set of all possible *percept sequences* P^* to the *set of actions* A formulated as an abstract mathematical function.



The **agent program** is a concrete implementation of this function for a given physical system.

Agent = architecture (hardware) + agent program (implementation of f)

- Sensors
- Memory
- Computational power

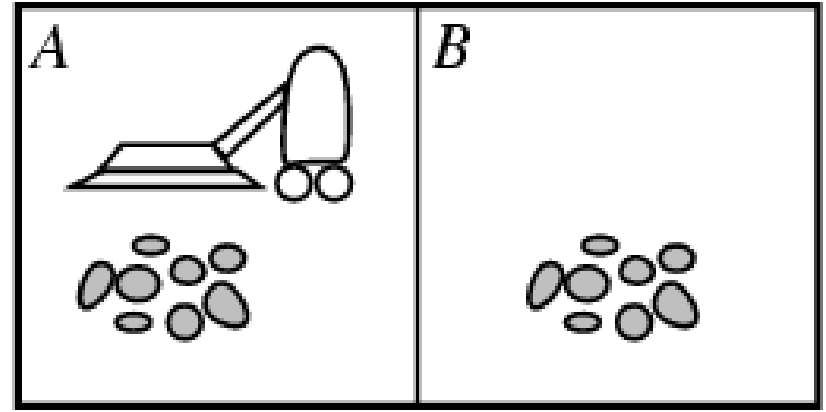
Example: Vacuum-cleaner World

- **Percepts:**

Location and status,
e.g., [A, Dirty]

- **Actions:**

Left, Right, Suck, NoOp



Most recent
Percept p

Agent function: $f : P^* \rightarrow A$

Percept Sequence	Action
[A, Clean]	Right
[A, Dirty]	Suck
...	
[A, Clean], [B, Clean]	Left
...	
[A, Clean], [B, Clean], [A, Dirty]	Suck
...	

Implemented agent program:

```
function Vacuum-Agent( [location, status] )  
    returns an action  $a$   
  
    if status = Dirty then return Suck  
    else if location = A then  
        return Right  
    else if location = B then  
        return Left
```

Problem: This table can become infinitively large!

Rational Agents

Rule: Pick the action that maximize the expected utility

$$a = \operatorname{argmax}_{a \in A} E(U \mid a)$$

This means:

- **Rationality is an ideal** – it implies that no one can build a better agent
- **Rationality $\neq$ Omniscience** – rational agents can make mistakes if percepts and knowledge do not suffice to make a good decision
- **Rationality $\neq$ Perfection** – rational agents maximize **expected** outcomes not actual outcomes
- **It is rational to explore and learn** – i.e., use **percepts** to supplement prior knowledge and become autonomous
- **Rationality is often bounded** by available memory, computational power, available sensors, etc.

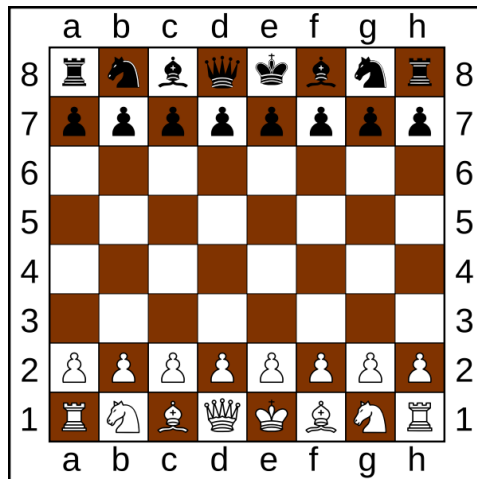
Environment Types: Observability

Fully observable: The agent's sensors give it access to the complete state of the environment. The agent can “see” the whole environment.

VS.

Partially observable: The agent cannot see all aspects of the environment. E.g., it can't see through walls

Examples:



Chess

- All Pieces visible
- Full information



Poker

- Some cards are not known (e.g., other player's cards)

Environment Types: Stochasticity

Deterministic: The environment follows fixed rules. Actions have always the same results. **vs.**

Stochastic: The environments reacts to actions with some randomness. The same action may have different results when tried multiple times.

Examples:



Tic-tac-toe

- The game follows fixed rules without uncertainty



Games with uncertainty:

- Roll dice to move
- Pick a card from a stack

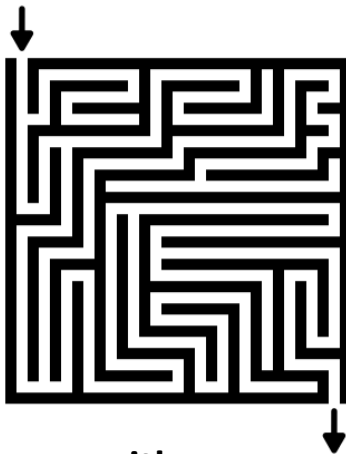
Environment Types: Knowledge

Known: The agent knows the rules of the environment and can predict the (chance of the) outcome of actions.

vs.

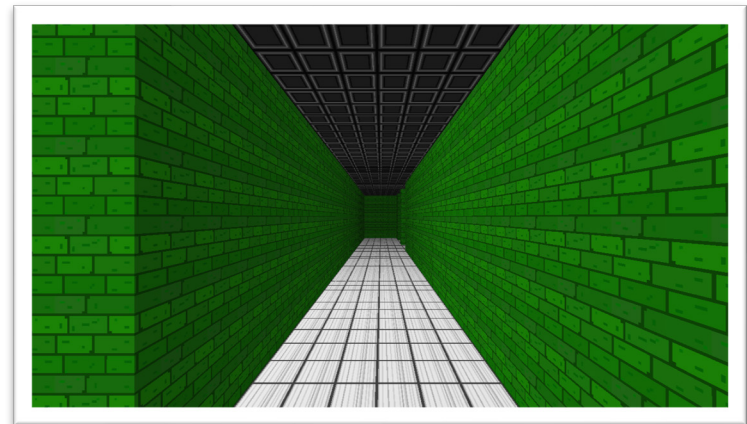
Unknown: The agent does not know the rules of the environment and cannot predict the outcome of actions.

Examples:



Solving a maze **with** a map

- The agent knows in every position where actions (North, East, etc) lead.
- The map can be used for planning.



Solving a maze **without** a map

- The agent does not know where all the corridors lead.
- The agent needs to explore.

The background is a dark, abstract composition featuring horizontal motion blur streaks in shades of blue, teal, and orange. In the lower right, there are curved, concentric lines that suggest a road or a tunnel, creating a sense of depth and movement.

Case Study: Self-Driving Cars

Self-driving Cars

SAE Automation Levels

- Level 1 - Driver Assistance (“hands on”)
- Level 2 - Partial Automation (“hands off”)
- Level 3 - Conditional Automation
- Level 4 - High Automation
- Level 5 - Full Automation (“steering wheel optional”)

Components

- Sensing
- Maps
- Path planning
- Controlling the vehicle

Why is this so hard?





A Self-Driving Car as a Rational Agents

Rule: Pick the action that maximize the expected utility

$$a = \operatorname{argmax}_{a \in A} \mathbb{E}(U \mid a)$$

Answer the following questions:

- If we have two cars and one provides more (expected) utility. Which car is rational?
- Can a rational self-driving car be involved in an accident?
- How would a self-driving car explore and learn?
- What does bounded rationality mean for a self-driving car?

PEAS Description of the Environment of a Self-Driving Car



Complete the PEAS description.

Performance measure	Environment	Actuators	Sensors

PEAS Description of the Environment of a Self-Driving Car



Performance measure	Environment	Actuators	Sensors
• Safe • fast • legal • comfortable trip • maximize profits	• Roads • other traffic • pedestrians • customers	• Steering wheel • accelerator • brake • signal • horn	• Cameras • sonar • speedometer • GPS • Odometer • engine sensors • keyboard

Environments for Self-Driving Cars



What properties does the environment have?

Fully observable: The agent's sensors give it access to the complete state of the environment. The agent can “see” the whole environment.

vs.

Partially observable: The agent cannot see all aspects of the environment. E.g., it can't see through walls

Deterministic: The environment follows fixed rules. Actions have always the same results.

vs.

Stochastic: The environments reacts to actions with some randomness. The same action may have different results when tried multiple times.

Known: The agent knows the rules of the environment and can predict the (chance of the) outcome of actions.

vs.

Unknown: The agent does not know the rules of the environment and cannot predict the outcome of actions.

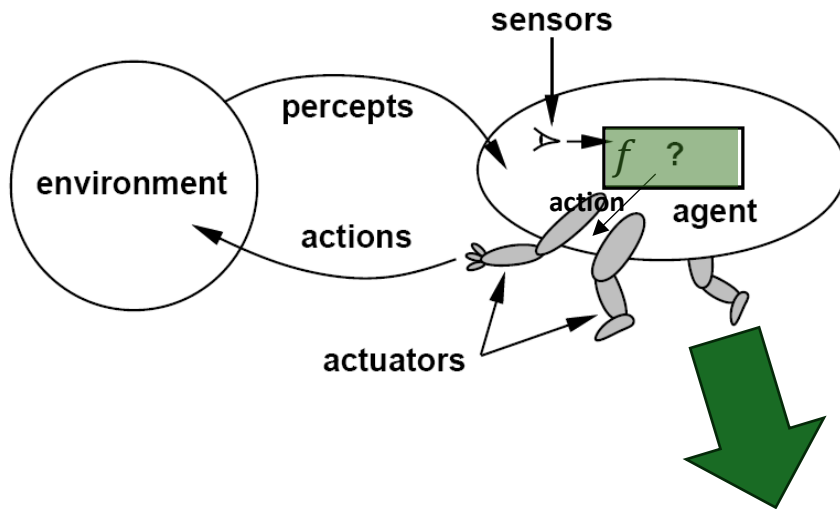
Assignment 1

Introduction : Environment

A white humanoid robot, ASIMO, stands in a dimly lit living room. The robot has a friendly face with large, dark eyes and a small smile. It is wearing a white suit with "ASIMO" printed on the chest. In the background, there is a wooden coffee table with a red cushion, a potted plant, and a television displaying the ASIMO logo. The room has warm lighting and framed pictures on the wall.

Module Review 2

Designing a Rational Agent

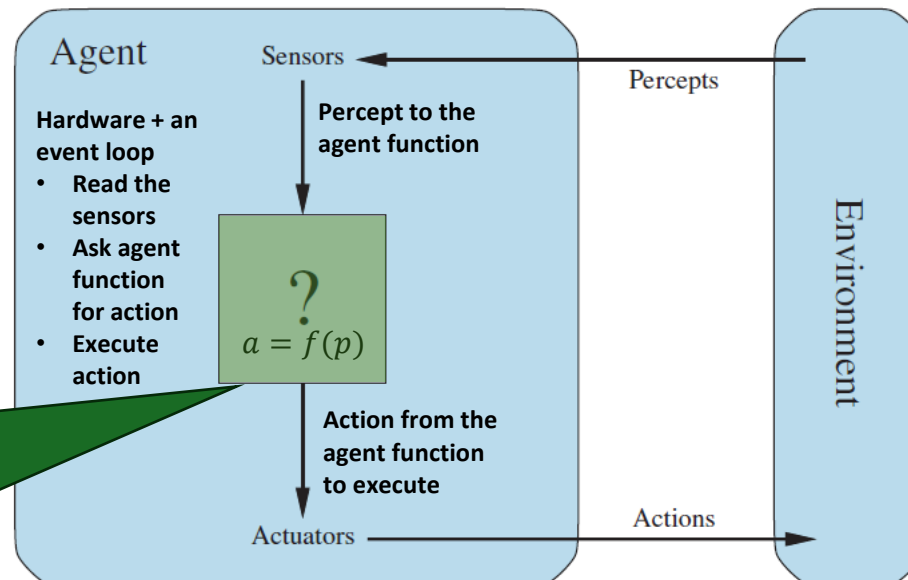


Remember the definition of a rational agent:

*“For each possible percept sequence, a rational agent should select an **action** that **maximizes its expected performance measure**, given the evidence provided by the **percept sequence** and the **agent’s built-in knowledge**.”*

Agent Function

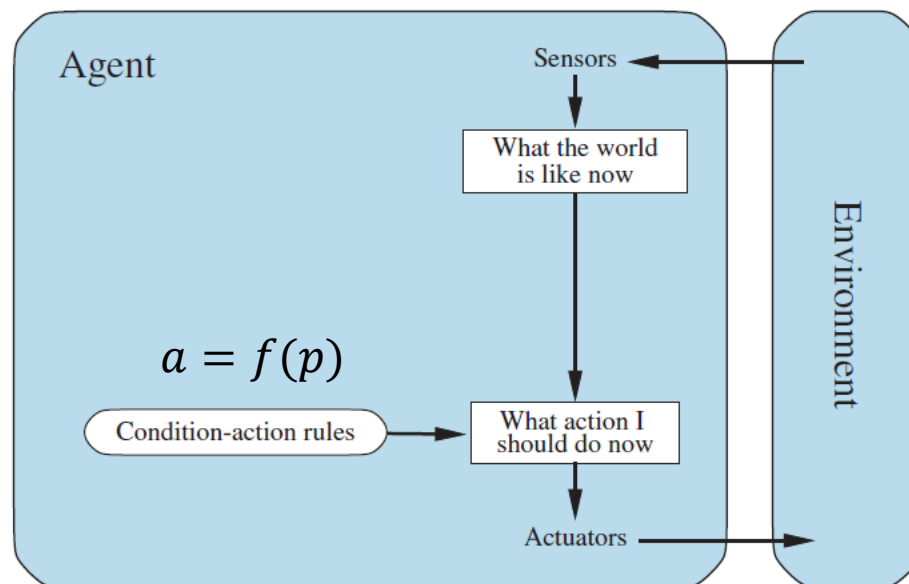
- Represents the “brain”
- Assess performance measure
- Remember percept sequence
- Built-in knowledge



Important: Everything outside the agent function represents the environment. This includes the physical robot, its sensors and its actuators, and event loop!

Simple Reflex Agent

- Uses only built-in knowledge in the form of **rules** that select action only **based on the current percept**. This is typically very fast!
- The **agent does not know about the performance measure**! But well-designed rules can lead to good performance.
- The agent needs no memory and ignores all past percepts.

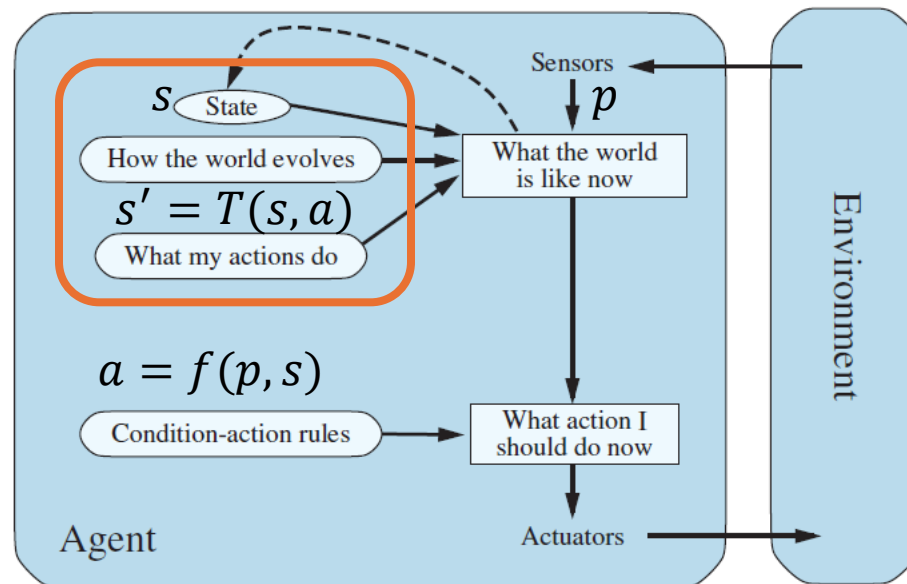


The interaction is a sequence: $p_0, a_0, p_1, a_1, p_2, a_2, \dots, p_t, a_t, \dots$

Example: A simple vacuum cleaner that uses rules based on its current sensor input.

Model-based Reflex Agent

- Maintains a **state variable** to keep track of aspects of the environment that cannot be currently observed. I.e., it has memory.
- It knows how the environment evolves over time given its last action. It updates the state using a **transition function** and the new percept.
- There is now more information for the **rules** to make better decisions.



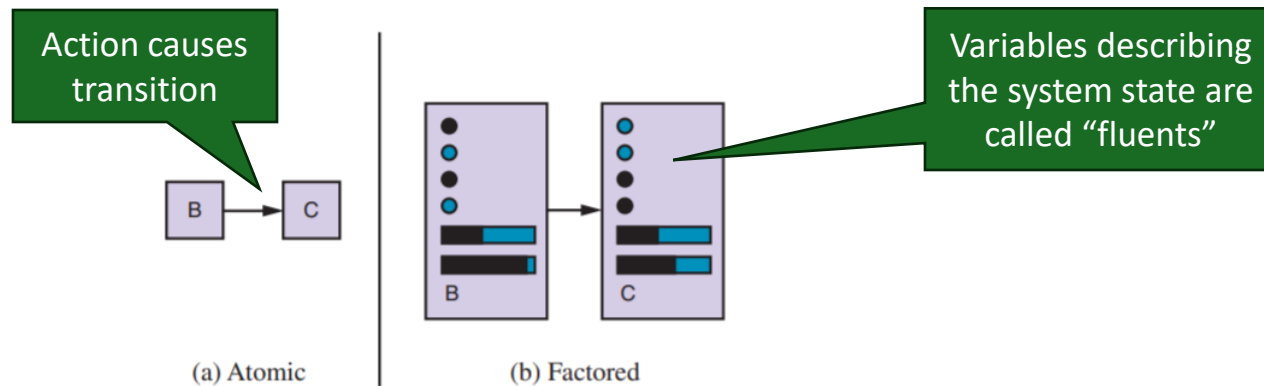
The interaction is a sequence: $p_0, s_0, a_0, p_1, s_1, a_1, p_2, s_2, a_2, p_3, \dots, p_t, s_t, a_t, \dots$

Example: A vacuum cleaner that remembers where it has already cleaned.

State Representation

States help to keep track of the environment and the agent in the environment. This is often also called the **system state**. The representation can be

- **Atomic**: Just a label for a black box. E.g., A, B
- **Factored**: A set of attribute values called fluents.
E.g., [location = left, status = clean, temperature = 75 deg. F]

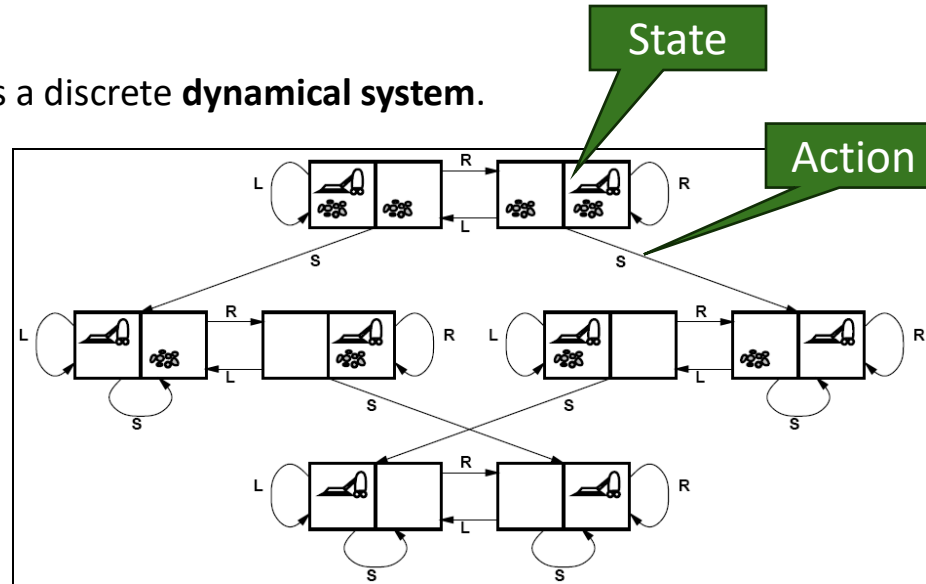


We often construct atomic labels from factored information. E.g.: If the agent's state is the coordinate $x = 7$ and $y = 3$, then the atomic state label could be the string $"(7, 3)"$. With the atomic representation, we can only compare if two labels are the same. With the factored state representation, we can reason more and calculate the distance between states!

The set of all possible states is called the **state space S** . This set is typically very large!

Transition Function

- The environment is modeled as a discrete **dynamical system**.
- Example of a state diagram for the Vacuum cleaner world.



- States change because of
 - a. System dynamics of the environment (the environment evolves by itself).
 - b. The actions of the agent.
- Both types of changes are represented by the transition function written as

$$T: S \times A \rightarrow S$$

or

$$s' = T(s, a)$$

S ... set of states
 A ... set of available actions
 $a \in A$... an action
 $s \in S$... current state
 $s' \in S$... next state

The background is a dark, abstract composition featuring horizontal motion blur streaks in shades of blue and orange. In the lower right, there are curved, concentric lines that suggest a road or a tunnel, creating a sense of depth and movement.

Case Study: Self-Driving Cars

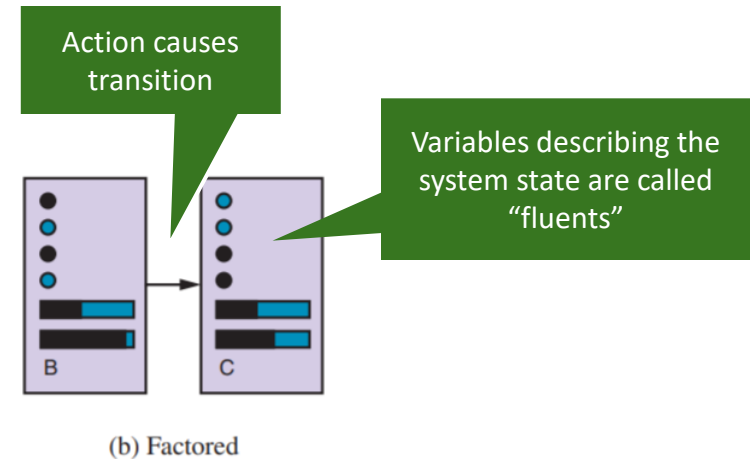


State Representation: Self-Driving Car

States help to keep track of the environment and the agent in the environment.

Design a structured representation for the state of a self-driving car.

- a) What fluents should it contain?
- b) What actions can cause transitions?
- c) Draw a small transition diagram.





Environment for a Self-Driving Car

- | | | |
|--|------------|--|
| ☐ Fully observable: The agent has access to the complete current state of the environment.
Has deterministic percepts that are 100% reliable. | vs. | ☐ Partially observable: The agent's sensors provide incomplete or noisy information about the state of the environment.
Noisy information means unreliable stochastic percepts (aka a stochastic sensor model) |
| ☐ Deterministic:
Deterministic transition function: Changes in the environment are completely determined by the current state of the environment and the agent's action. | vs. | ☐ Stochastic:
Stochastic transition function: leads to transition probabilities and a Markov process. |
| ☐ Known: The agent knows the transition function . | vs. | ☐ Unknown: The needs to learn the transition function by trying actions. |
| ☐ Static: The environment is not changing while agent is deliberating. | vs. | ☐ Dynamic: The environment is changing while the agent is deliberating. |



Check what applies and explain what it means for a self-driving car.



What Type of Intelligent Agent is a Self-Driving Car?

☐ Is it learning?

☐ Utility-based agents

☐ Goal-based agents

☐ Model-based reflex agents

☐ Simple reflex agents

Does it collect utility over time? How would the utility for each state be defined?

Does it have a goal state?

Does it store state information. How would they be defined (atomic/factored)?

Does it use simple rules based on the current percepts?



Check what applies

Why is this so hard?

- Self-driving cars operate in a very complicated partially observable, stochastic, and dynamic environment.
- Can only use bounded rationality because of limits with sensors and computational power.
- Require a set of different agents that cooperate.





What You Should Know

- What an **agent function**
 $action = f(percepts)$
is and how it interacts with the environment.
- What are **states** and what is the **transition function**?
- How **environments** differ in terms of observability, uncertainty (stochastic behavior), and if the transition function is known.
- How to identify different **types of agents**.

Assignment 1

Reflex Agents